

2010 Flash Crash E-Mini Futures MIS Case

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Abstract

The May 6, 2010 “flash crash” shows how a large, seemingly routine algorithmic order in stock index futures can interact with fragile market conditions and high-frequency trading to create a sudden, system-wide breakdown in prices. A major institutional investor, widely identified as Waddell & Reed, initiated a program to sell 75,000 E-Mini S&P 500 futures contracts (about \$4.1 billion notional) as a hedge, using an automated strategy that keyed off trading volume rather than price.¹ As this sell program hit already volatile markets, high-frequency traders and cross-market arbitrageurs rapidly recycled contracts among themselves and transmitted selling pressure into related stocks and ETFs, driving an extreme but short-lived plunge and rebound in prices.² This case highlights how algorithm design, market microstructure, and risk controls jointly determine the resilience or fragility of highly automated financial systems.

1 Background: Waddell & Reed and E-Mini Futures

Waddell & Reed Financial, Inc. was a long-established U.S. asset management and financial planning company, tracing its history to 1937 and known for managing mutual funds for retail and institutional investors.³ Like many institutional investment firms, Waddell & Reed made money by managing other people’s portfolios—pension funds, university endowments, and individual retirement accounts. The firm positioned itself as a traditional, fundamentals-based investor, running large equity portfolios from its Kansas City base.⁴ They analyzed companies’ financials and held stocks for the long term, not the millisecond trades that high-frequency firms were beginning to dominate.

The E-Mini S&P 500 futures contract is a standardized, electronically traded agreement on the Chicago Mercantile Exchange (CME) that allows investors to gain or hedge exposure to the S&P 500 stock index.⁵ Each contract’s value moves in lockstep with the broad

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¹[SEC/CFTC]: <https://www.sec.gov/news/studies/2010/marketevents-report.pdf>

²[Wikipedia–Flash Crash]: https://en.wikipedia.org/wiki/2010_flash_crash

³[Waddell & Reed History]: <https://www.fundinguniverse.com/company-histories/waddell-reed-inc-history/>

⁴[Wikipedia–Waddell & Reed]: https://en.wikipedia.org/wiki/Waddell_&_Reed

⁵[Schwab–E-Mini S&P 500]: <https://www.schwab.com/futures/sp-500-emi>

U.S. stock market, making it useful for hedging—essentially, buying insurance against market declines.⁶ If you own \$100 million in stocks and worry about a crash, you can sell E-Mini futures. If stocks fall, your portfolio loses value but the futures gain, offsetting the damage. Because E-Mini futures trade electronically at high speed, they sit at the core of modern index trading, tightly linked to ETFs like SPY and the 500 underlying stocks through arbitrage.

2 Market Conditions Before the Crash

May 6, 2010 began with elevated volatility. European sovereign debt concerns—particularly Greece’s financial crisis—had markets on edge.⁷ Liquidity was thin. In normal conditions, you can trade large positions without moving prices much because there are plenty of buyers and sellers. That day, order books were fragile.⁸ Fewer participants meant even moderate orders could swing prices.

High-frequency traders were active in both E-Mini futures and related products, constantly arbitraging tiny price differences.⁹ These algorithmic traders buy and sell in milliseconds, acting as short-term market makers but quick to pull back when volatility spikes.¹⁰ That hair-trigger behavior would matter.

3 The Algorithm and What Went Wrong

That afternoon, Waddell & Reed needed to hedge a large equity portfolio against further market declines. Rather than selling stocks directly—which would require hundreds of transactions across different securities—they chose to sell 75,000 E-Mini contracts through a broker’s automated algorithm.¹¹ The order represented roughly \$4.1 billion notional (meaning the total market value it controlled) and about 9% of that day’s E-Mini volume.¹²

Here’s where design mattered. The algorithm was volume-based: it aimed to sell contracts as a fixed percentage of overall market trading volume.¹³ If the market was trading 10,000 contracts per minute, the algorithm might sell 1,000. If volume jumped to 20,000, it would accelerate to 2,000. The problem? It ignored price.¹⁴

⁶[Dorman Trading—Contract Specs]: <https://www.dormantrading.com/TraderTools/CMEeminisp500.pdf>

⁷[SEC/CFTC]: <https://www.sec.gov/news/studies/2010/marketevents-report.pdf>

⁸[Wikipedia—Flash Crash]: https://en.wikipedia.org/wiki/2010_flash_crash

⁹[UCSC Flash Crash Overview]: <https://news.ucsc.edu/2016/03/flash-crash/>

¹⁰[MarketWatch—SEC, CFTC blame algorithm]: <https://www.marketwatch.com/story/sec-cftc-blame-algorithm-for-flash-crash-2010-10-01-1246290>

¹¹[NYT—Lone \$4.1 Billion Sale]: <https://www.nytimes.com/2010/10/02/business/02flash.html>

¹²[SEC/CFTC]: <https://www.sec.gov/news/studies/2010/marketevents-report.pdf>

¹³[SEC/CFTC]: <https://www.sec.gov/news/studies/2010/marketevents-report.pdf>

¹⁴[AI-CIO Analysis]: <https://www.ai-cio.com/news/flash-crash-report-points-finger-at-wrong-source-of-big-trade-that-went-bad/>

This created a feedback loop. As the algorithm sold, prices fell. Falling prices made traders nervous, increasing volume. Higher volume triggered faster selling, pushing prices lower still, which created more volume. The algorithm was essentially accelerating into its own wake.

Meanwhile, high-frequency traders bought contracts from the algorithm and immediately tried to offload them, creating a “hot potato” effect.¹⁵ Contracts bounced between HFT firms rather than finding stable buyers. Each trade counted as volume, further inflating the signal that told Waddell & Reed’s algorithm to sell faster.

Cross-market arbitrageurs noticed E-Mini prices falling and automatically sold stocks and ETFs to maintain price alignment.¹⁶ This transmitted stress from futures into equities. As volatility spiked, many algorithmic market makers withdrew entirely, draining liquidity just when it was needed most.¹⁷

Within minutes, stocks traded at absurd prices—pennies or \$100,000—before snapping back.¹⁸ The Dow fell nearly 1,000 points and recovered most of it in roughly 20 minutes. Once Waddell & Reed’s program finished and traders realized nothing fundamental had changed, prices returned to normal.

4 Aftermath and Lessons

The SEC/CFTC report identified the volume-based sell program as a key trigger, though not the sole cause.¹⁹ It interacted with thin liquidity, high-frequency trading, and tightly coupled markets to produce a systemic event. Waddell & Reed argued it had followed standard procedures using a common broker algorithm.²⁰ Who’s responsible when many algorithms, each following reasonable rules, collectively produce chaos?

Regulators responded with circuit breakers—trading pauses triggered by rapid price moves—and limit up/limit down rules that prevent stocks from trading outside reasonable bands.²¹ These mechanisms give humans time to assess situations and break runaway feedback loops.

The core lessons: algorithm design choices matter enormously. A price-sensitive algorithm would have slowed as prices fell, breaking the cycle. System interconnections amplify shocks—what starts in futures spreads instantly to stocks and ETFs. And real-time controls are essential when automated systems can spiral faster than humans can react.²²

¹⁵[Wikipedia–Flash Crash]: https://en.wikipedia.org/wiki/2010_flash_crash

¹⁶[Econbrowser–Causes of the Flash Crash]: https://econbrowser.com/archives/2010/10/causes_of_the_f

¹⁷[UCSC Flash Crash Overview]: <https://news.ucsc.edu/2016/03/flash-crash/>

¹⁸[The Economist–One big, bad trade]: <https://www.economist.com/newsbook/2010/10/01/one-big-bad-trade>

¹⁹[SEC/CFTC]: <https://www.sec.gov/news/studies/2010/marketevents-report.pdf>

²⁰[Reuters–Waddell & Reed says not cause]: <https://www.reuters.com/article/world/waddell-reed-says-not-cau-se-of-may-6-flash-crash-idUSTRE64D42W/>

²¹[Wikipedia–Flash Crash]: https://en.wikipedia.org/wiki/2010_flash_crash

²²[Strategy+Business–Predicting Flash Crashes]: <https://www.strategy-business.com/article/00144>

5 Questions

1. **BUSINESS-TECHNOLOGY ALIGNMENT: Understanding how risk-management goals translate into algorithmic trading strategies.**
 - Waddell & Reed used E-Mini futures to hedge a large equity portfolio rather than selling stocks directly. How did this decision align with its business and risk-management strategy, and what does it reveal about the role of complex market infrastructure in implementing seemingly simple hedging objectives?
2. **ALGORITHM DESIGN AND MARKET SENSITIVITY: Recognizing the impact of design choices in automated execution strategies.**
 - The sell program was based largely on trading volume, not on price movements. Why is this a critical design choice from an MIS perspective, and how might a more price-sensitive or adaptive algorithm have changed the outcome on May 6, 2010?
3. **INTERCONNECTED SYSTEMS AND FEEDBACK LOOPS: Understanding how tightly coupled markets can amplify local shocks.**
 - The flash crash involved E-Mini futures, ETFs, individual stocks, and high-frequency traders acting across markets. How did the interconnections between these systems contribute to the speed and severity of the crash, and what parallels can you draw to complex information systems in other industries?
4. **RISK CONTROLS AND CIRCUIT BREAKERS: Evaluating preventive and detective controls in real-time systems.**
 - After the flash crash, regulators introduced measures such as limit up/limit down rules and wider use of trading pauses. If you were designing risk controls for a real-time trading platform, what combination of technical safeguards (e.g., kill switches, anomaly detection) and organizational processes (e.g., escalation protocols) would you put in place?
5. **ACCOUNTABILITY AND GOVERNANCE: Considering responsibility in shared, algorithmic environments.**
 - In this case, responsibility was spread across a mutual fund (Waddell & Reed), a broker providing the algorithm, high-frequency traders, and market operators. How should accountability be assigned when a system-wide event emerges from the interaction of many algorithms, and what governance structures could improve transparency and oversight?